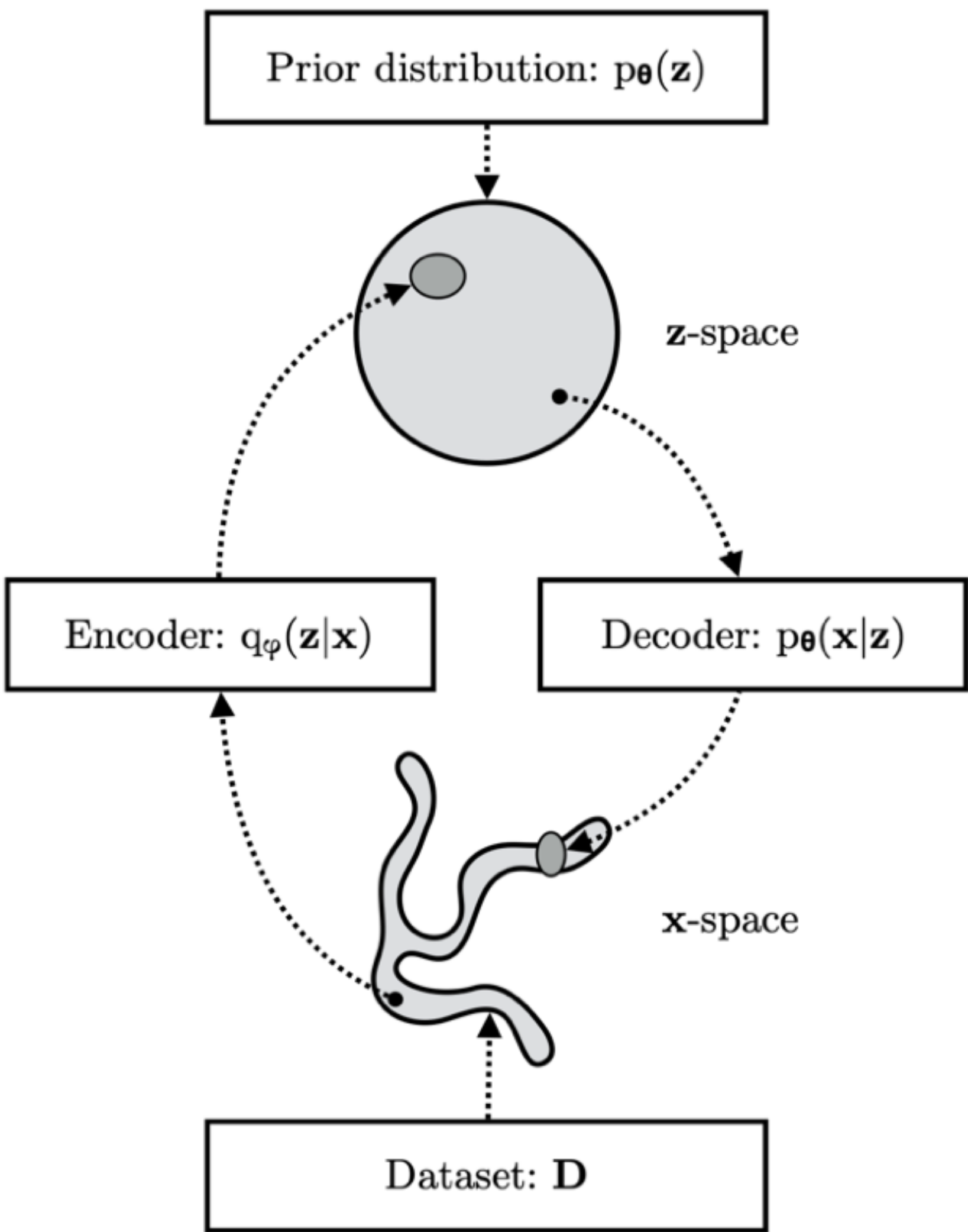


Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

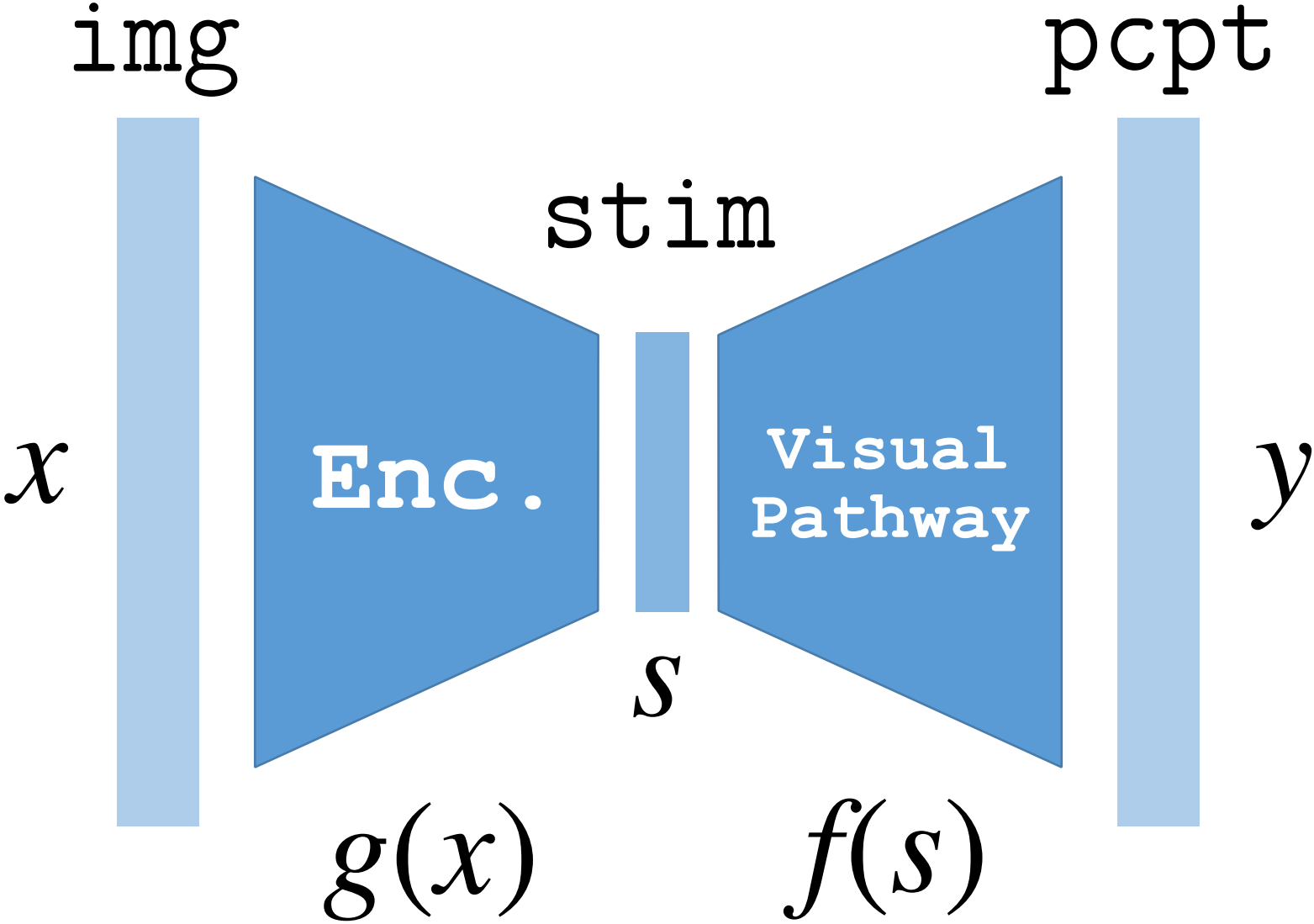
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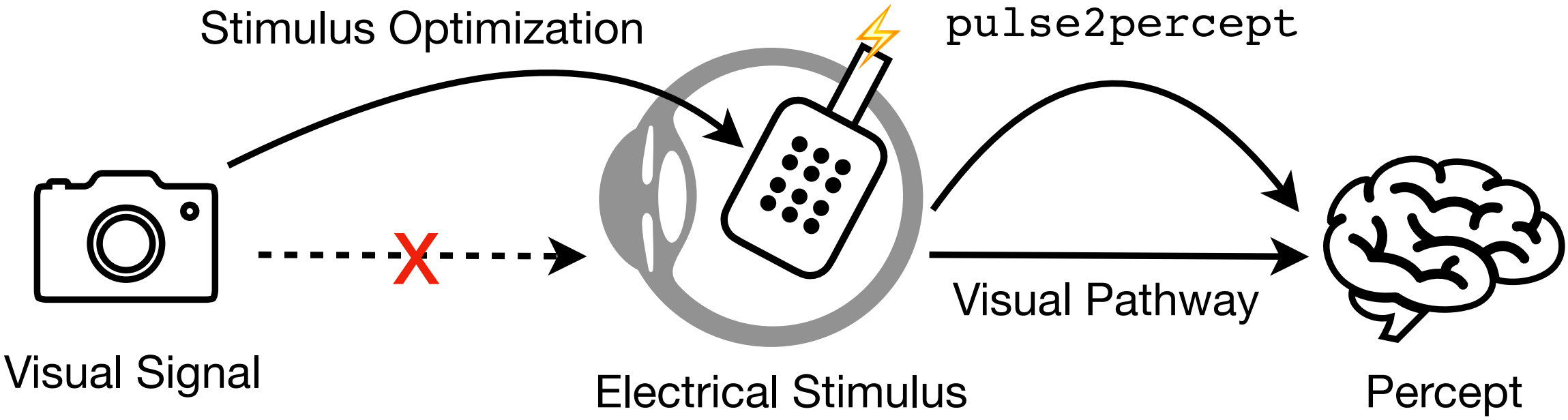


Investible Encoding

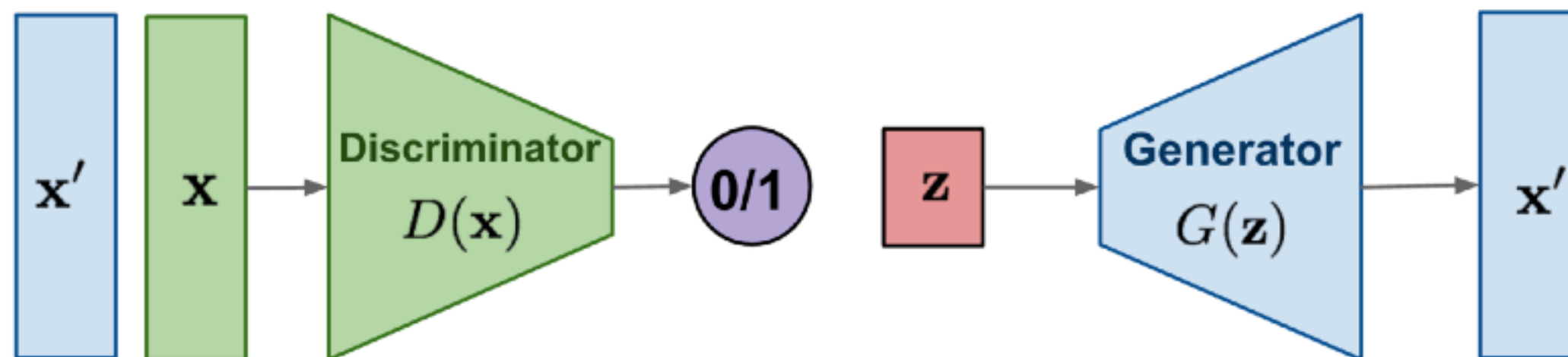


[Kingma, PhD Thesis, 2017]

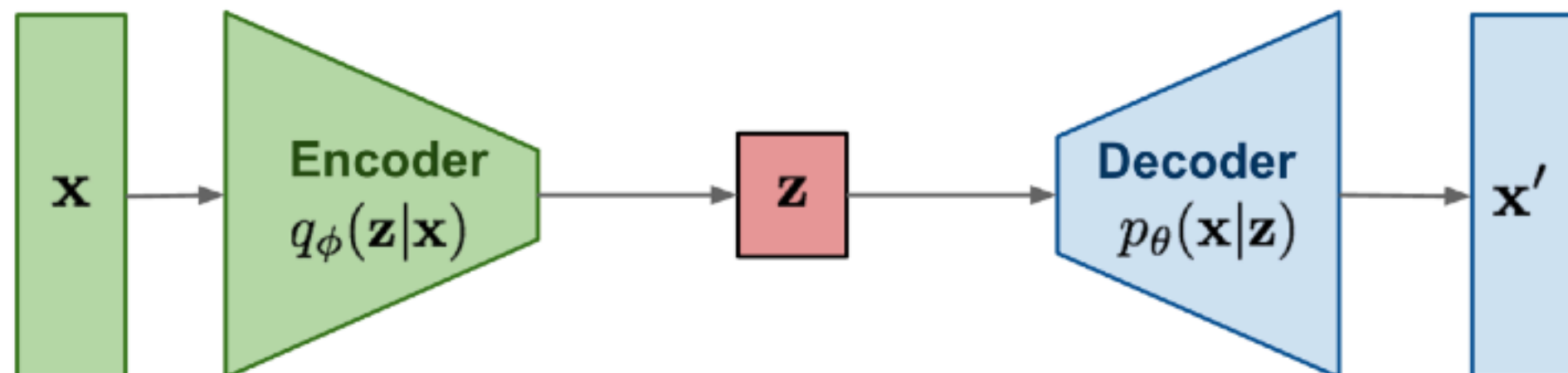




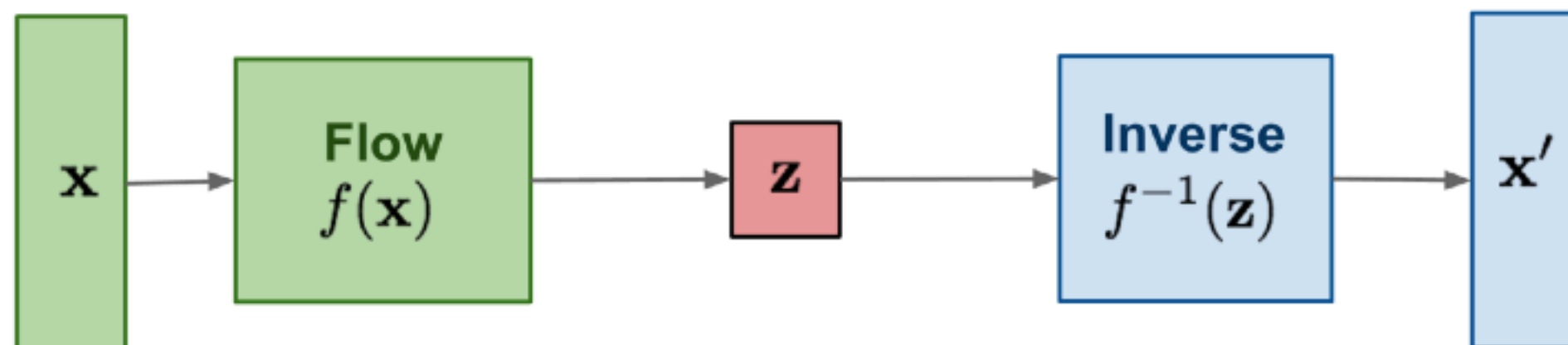
GAN: Adversarial training



VAE: maximize variational lower bound



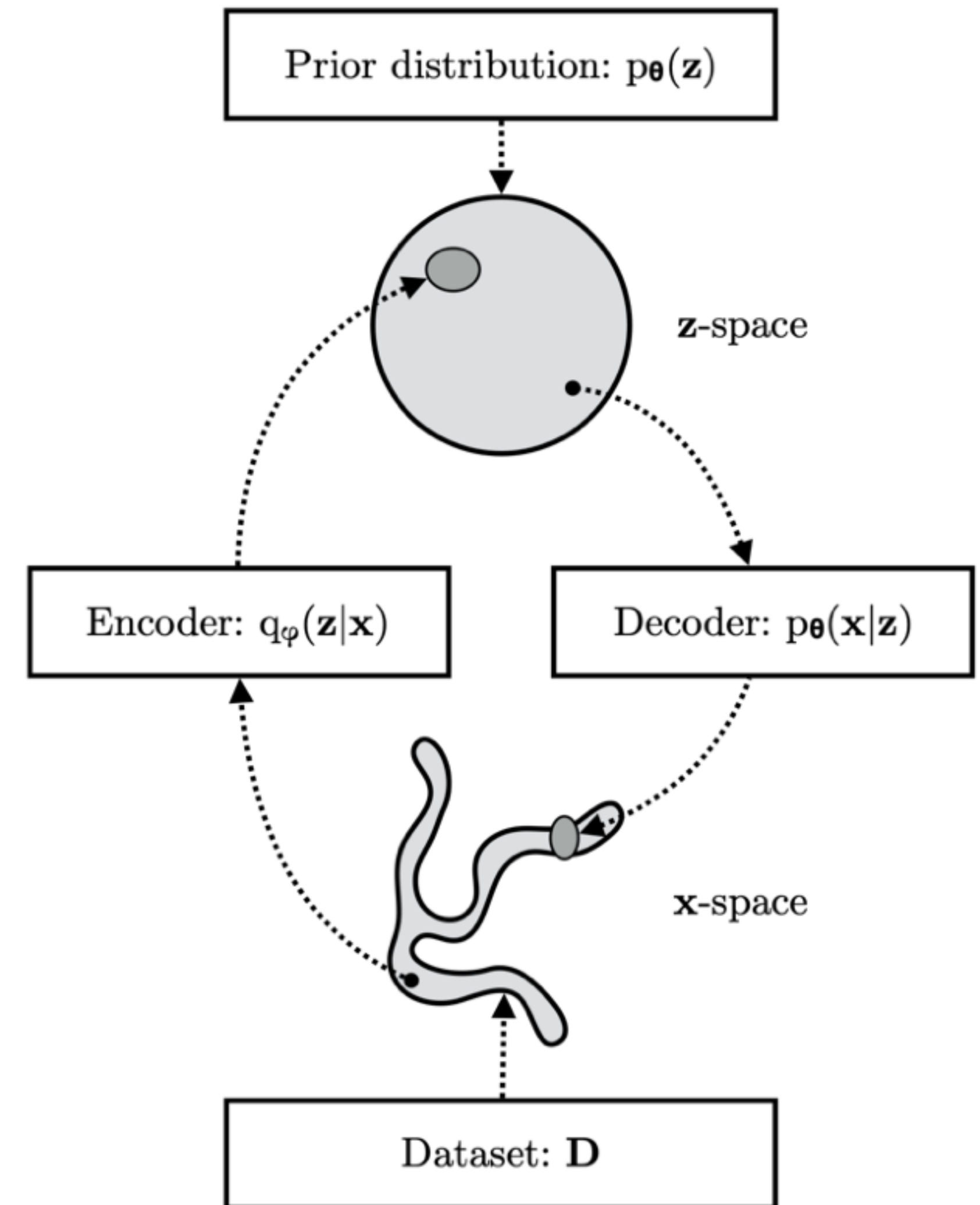
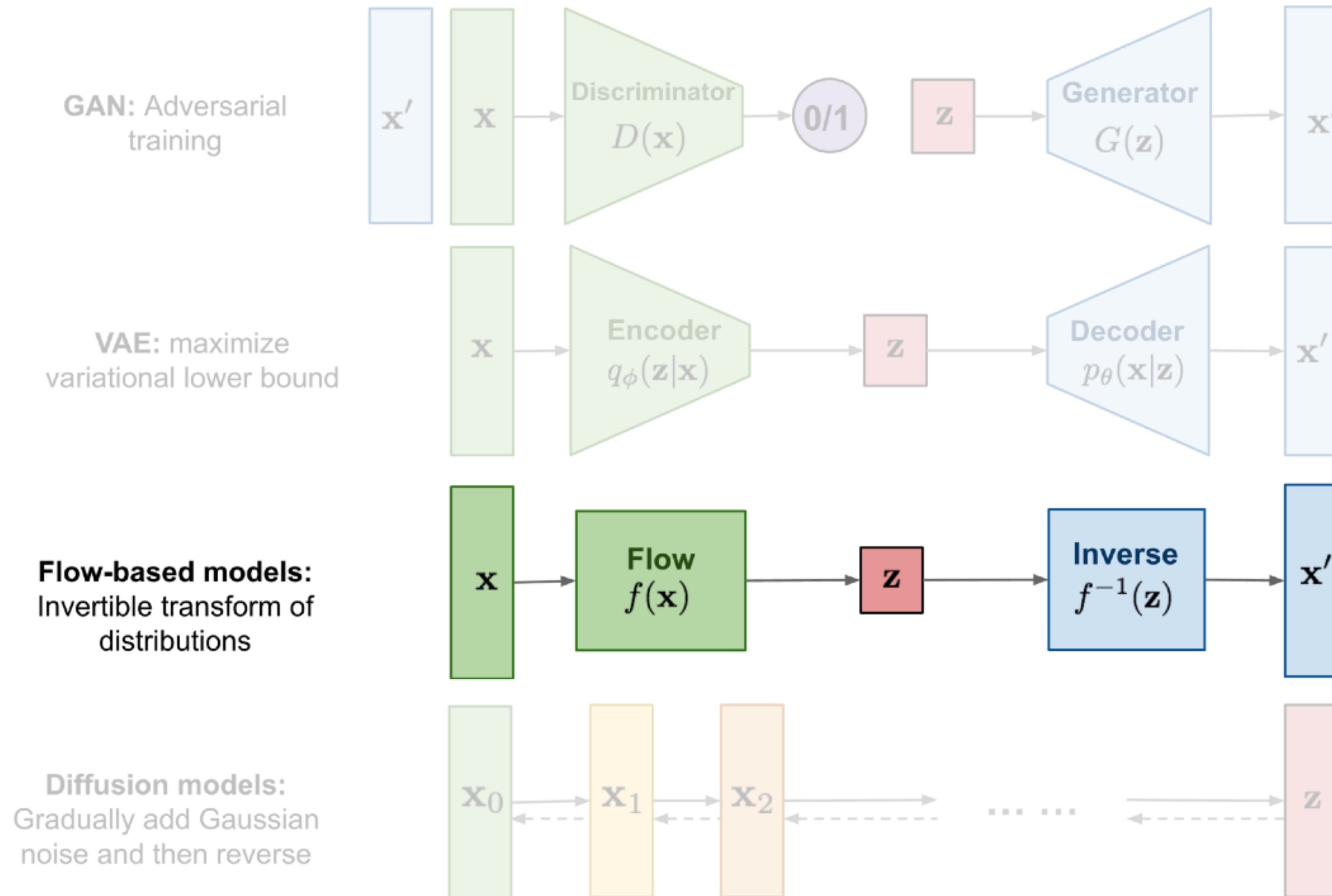
Flow-based models:
Invertible transform of distributions



Diffusion models:
Gradually add Gaussian noise and then reverse



Invertible Encoding



<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

[Kingma, PhD Thesis, 2017]

Invertible Neural Networks

Likelihood: $\mathcal{L}(\theta \mid x) = p_{\theta}(x) = P_{\theta}(X = x)$

Probability of observing data x , assuming θ is the actual parameter.

$$p_{\theta}(x) = \int p_{\theta}(x, z) dz \quad \text{Intractable due to "combinatorial explosion" of } z$$

